

SATNET EDU / Lecture 04

Dynamics, Handover, and Performance

How changing links and routes affect communication over time.

LECTURE 04

From a working snapshot to a changing connection

Follow the timeline. Explain the transition. Measure the effect.

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A Working Route Is a Statement About One Time

Satellite motion changes link availability and distance even when both ground terminals stay still.

► **Edges can appear or disappear.**

- A terminal crosses a footprint edge. An ISL exceeds its range. A scheduled failure disables a node.

► **Weights can change without an outage.**

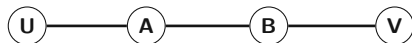
- The same sequence of satellites can remain connected while changing distances alter its propagation delay.

► **Track the graph through time.**

- A route at time t does not prove that the same path will work at $t + \Delta t$.

Same endpoints, a different usable chain

Time t_0



Time t_1



U now enters through B

Constructed snapshots. Dashed lines mark an unavailable former link. No control-plane timing is implied.

Handover Transfers Service Between Access Links

Overlapping contact windows provide time to prepare a new access link before the old one ends.

► **Separate availability from selection.**

- A terminal may see both A and B, while its traffic uses only A. Visibility does not specify a handover policy.

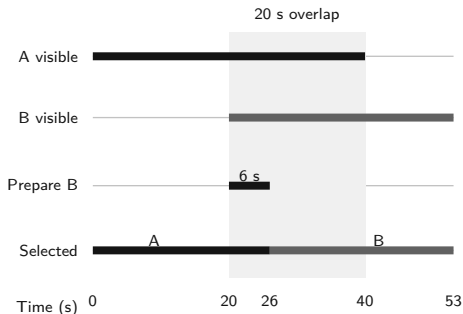
► **Use the overlap to prepare.**

- Acquire the new link, validate it, and transfer traffic before releasing the old link when hardware permits.

► **Account for setup time.**

- A 20-second overlap with 6 seconds of preparation leaves at most 14 seconds for other actions and margin.

An overlap creates a preparation opportunity



Constructed make-before-break example. Dual-link preparation requires supporting terminal hardware.

The Best Link Now May Leave Soon

An access policy can consider both current geometry and the time remaining before contact ends.

► **High elevation can shorten the link.**

- It often gives a shorter slant range. A descending satellite can still be close to the end of its contact.

► **Remaining contact time is different.**

- A rising satellite may have a longer usable future window even when its current elevation is lower.

► **Avoid switching on every small gain.**

- A hysteresis margin requires a meaningful improvement. A minimum dwell time can also reduce repeated switching.

Two candidates at the same decision time

Candidate	Elevation	Time left
A, descending	55°	35 s
B, rising	40°	180 s

Best elevation now
chooses A

Longer contact
chooses B

Example rule: switch only if the new score exceeds the old score by margin H

Illustrative candidate values and policies. Future visibility does not guarantee capacity or route availability.

Path Changes Can Make Delay Jump

Orbital motion changes a path smoothly, but selecting a different path can cause a sudden shift.

► **Follow each candidate path.**

- Its propagation cost varies as the satellites move. Availability can end before its cost becomes unattractive.

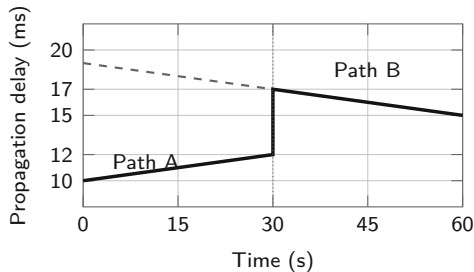
► **A switch can add delay.**

- Here the 12 ms path disappears at 30 seconds. The available replacement costs 17 ms at that moment.

► **Report changes as well as averages.**

- Two traces with the same mean delay can have different jumps, interruptions, and route lifetimes.

The cheapest path becomes unavailable



Bold: selected path **Dashed:** available alternative

Constructed path costs. The jump is 5 ms. No packet loss or queueing is inferred from this plot.

Propagation Is Only One Part of Packet Delay

A short route through space can still carry packets slowly when other delay terms are large.

► Separate the four common terms.

- Propagation moves a signal.
Transmission places bits on a link.
Queueing waits for service. Processing examines the packet.

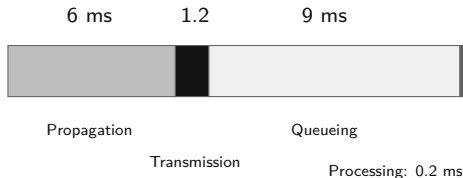
► Use compatible units.

- A 1500-byte packet has 12000 bits. On a 10 Mb/s link, transmission takes 1.2 ms before adding other terms.

► Do not label propagation as RTT.

- Round-trip time includes forward and return paths. An application may also wait for servers or additional exchanges.

One illustrative hop: 16.4 ms total



$$D = \sum_{\text{hops}} (d/c + L/C + q + p)$$

L : bits, C : bit/s, q : wait, p : processing

Constructed store-and-forward example. SatNet Edu computes only the propagation term $\sum d/c$.

An Outage Can Leave a Queue After Access Returns

Restoring a link stops the outage, but it does not instantly remove accumulated traffic.

► **Traffic can accumulate during a gap.**

- At an arrival rate of 20 Mb/s, a 0.5-second outage builds 10 Mbit if the buffer has enough space.

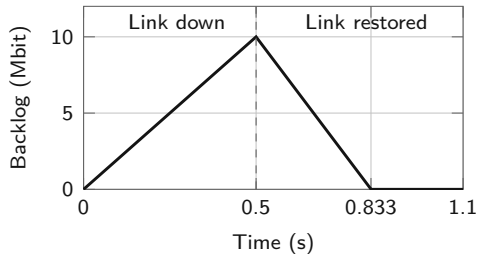
► **Drain using spare service rate.**

- After recovery at 50 Mb/s, ongoing arrivals still consume 20 Mb/s. Only 30 Mb/s remains to clear the backlog.

► **Buffer limits change the outcome.**

- A smaller buffer loses traffic instead. When service is no greater than arrivals, the queue cannot drain.

Recovery continues beyond link restoration



$$\text{Drain time} = 10 / (50 - 20) = 0.333 \text{ s}$$

Ideal fluid queue, constant arrivals, adequate buffer. This calculation is outside the SatNet Edu model.

A Sender Needs Enough Data in Flight

A sender waiting for acknowledgements can underuse a fast path when its send window is too small.

► **Relate capacity to feedback time.**

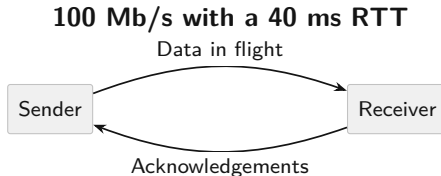
- At bottleneck rate C and round-trip time RTT , the bandwidth-delay product is $C \times RTT$ bits.

► **A small window can limit the rate.**

- With at most W unacknowledged bits, an ideal window-limited flow achieves no more than W/RTT .

► **RTT changes the required window.**

- At 100 Mb/s and 40 ms RTT , 4 Mbit must be in flight to keep the ideal path busy.



$$BDP = 100 \times 0.040 = 4 \text{ Mbit}$$

A 1 Mbit window gives at most
 $1/0.040 = 25 \text{ Mb/s}$ in this ideal model

Ideal continuous transfer. Congestion, loss, receiver limits, and protocol behavior can reduce throughput further.

A Faster New Path Can Reorder Packets

Packets sent later can arrive earlier when a flow moves to a shorter-delay path.

► **Compare send and arrival times.**

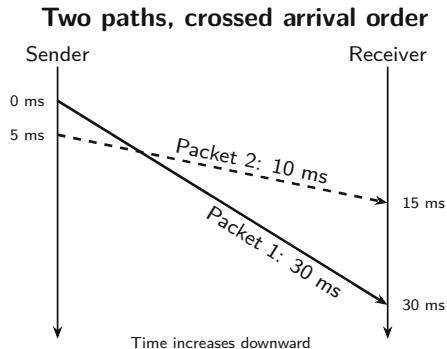
- Packet 1 leaves at 0 ms on a 30 ms path.
Packet 2 leaves at 5 ms on a 10 ms path.

► **Arrival order can reverse.**

- Packet 2 arrives at 15 ms, before packet 1 at 30 ms. A route change can reorder traffic without losing either packet.

► **Applications depend on timing.**

- Real-time playback needs timing margin.
Reliable transports may buffer out-of-order data before delivery.



Constructed packet-timing example. Serialization is omitted. SatNet Edu does not simulate packet arrivals.

A Coarse Recording Can Hide a Short Interruption

Recorded snapshots support claims about sample times, not every instant between them.

► State what was sampled.

- A 30-second recording can miss an outage entirely when the gap lies between two samples.

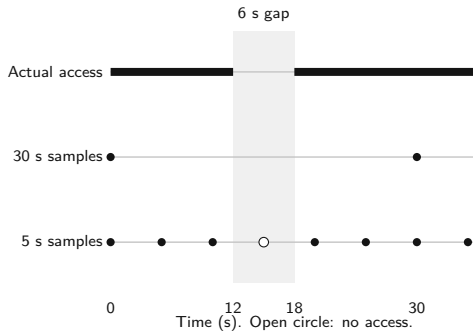
► Refine the observation interval.

- Compare 30-second and 5-second recordings using the same scenario. Check whether important transitions become visible.

► Keep model limits explicit.

- SatNet Edu inserts scheduled failure times. Changes caused by geometry still require sufficiently frequent samples.

A gap from 12 to 18 seconds



Constructed unscheduled gap. A 5-second interval detects this gap but still does not locate its exact start and end.